

Data Lakehouse Architecture

Document ID

DOC013

Project

Data Lakehouse

Category

Data Engineering

Batch

batch_4

Purpose: Original educational notes created as a searchable PDF corpus for a portfolio-scale incremental RAG pipeline. The material is designed to contain meaningful technical sections that naturally produce multiple retrieval chunks.

Lakehouse Overview

A data lakehouse combines the flexible storage characteristics of a data lake with management features commonly associated with data warehouses. Data can remain in scalable object storage while table formats and processing engines provide schema management, transactional updates, and reliable querying. The goal is to reduce unnecessary movement between separate lake and warehouse systems while supporting analytics and data engineering workloads on shared data.

Bronze, Silver, and Gold Layers

A common lakehouse design organizes data into Bronze, Silver, and Gold layers. Bronze contains raw or minimally processed source data. Silver contains cleaned, standardized, and deduplicated data that can be reused across workloads. Gold contains business-oriented tables or aggregates designed for reporting and analytics. These layers make data quality and transformation boundaries explicit without requiring every project to adopt exactly the same physical folder structure.

Reliable Table Management

Lakehouse table formats can add transactional behavior and metadata management on top of files stored in object storage. This improves reliability when pipelines perform inserts, updates, or incremental processing. Schema enforcement and schema evolution help control how table structures change over time. Versioned table history can also support debugging, auditing, and recovery when a pipeline writes incorrect data.

Incremental Processing

Large lakehouse pipelines should avoid repeatedly processing all historical data when only a small amount has changed. Incremental strategies may use timestamps, watermarks, change data capture, or source-specific identifiers to select new records. The transformation layer then merges or appends only the required data. This reduces compute usage and shortens processing time while still keeping curated tables current.

Serving Analytics

Gold-layer tables are commonly designed for downstream analytics. They may contain fact tables, dimensions, aggregates, or other business-ready structures. Data engineers should preserve lineage from raw inputs through transformations so users can understand where metrics originated. A lakehouse architecture is strongest when storage, processing, quality checks, and governance are treated as connected parts of the same data lifecycle.